Autonomous Perception Through Emergent Signs: The Conditions for an Emergent Division of Labour Among Independent Learning Agents

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Abstract

Multi-agent systems divide their work through roles, protocols and vocabularies that a designer fixes in advance. For populations that keep learning, the division of work has to come from the agents themselves, and there is no measured account, against a control, of the conditions under which it does. This paper is about the principles of social emergence that apply to communicating agents with autonomous perception: agents that each interpret signs through their own learning and never copy another's interpretation, whatever else they know. Its claim is that something forms socially, on top of anything the agents bring, and that it can be measured as the coevolution of an institution. An institution here is a system of roles held in place by what the participants expect of one another, which none of them was given. We study two small markets of program-building agents in which an agent perceives another only through a sign the other has learned to display and chooses partners only through a sign it has learned to seek, each agent learning on its own reward. The test of emergence is that a population that starts homogeneous differentiates through interaction into distinct, interlocking roles that none were assigned. It is measured by the mutual information between signs and what the agents do, against a shuffled null, and every run is paired with a sign-blind twin on the same seed. With signs, buyers obtained more of what they wanted on 12 of 12 seed pairs (sign test $p = 2.4e-04$). Sellers sorted into roles that an outsider could tell apart from their signs alone in 10 of 12 worlds against 1 of 12 twins (Fisher's exact test $p = 3.2e-04$). Where no learner could build the whole alone, wholes were sold in 4 of 12 worlds with signs against 0 of 12 twins at 10,000 rounds ($p = 0.047$), every one assembled from bought parts. The roles and the assembling teams persisted under a turnover that replaced nearly every agent who had formed them, and so outlived the individuals. A fresh population seeded with a mature one's practices rebuilt at once what single agents do, and only through renewed coevolution what agents do together. What formed is an emergent institution in the sociologist's sense, held in what the agents expect of one another and in no one of them. Read criteria were declared before the runs, with one exception reported; every roles statistic carries a shuffled null; seeds are pinned and the records are released.

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Keywords

division of labour, emergent communication, signalling, multi-agent learning, evolution strategies, markets

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1 Introduction

In almost every multi-agent system the division of work is designed in advance. The roles the agents play, the protocol by which they talk, the way tasks are allocated among them and the vocabulary they use are given by the designer. This holds from the classical architectures to today's teams of language-model agents, in which an orchestrator assigns the roles [23, 43]. Where communication is learned rather than given [20, 27, 31], the channel and the task are still fixed, and the protocol is learned by a gradient shared across the agents. Designing the division of work stops being possible when the agents form a population that keeps learning, when tasks arrive as compositions of parts, and when the agents are language models whose coordination no one can script [2, 8, 32, 33].

What is wanted in those settings is a division of labour that emerges from the agents' own learning and holds without a designer: who makes what, who buys from whom, and the meanings by which the agents find one another. The study of multi-agent systems is also the study of what emerges when agents with their own goals communicate: how conventions, markets, roles and institutions come about from local interaction, under which conditions, and how an observer can tell a structure that emerged from one the designer put there. Three difficulties stand in the way. Populations whose members learn from one another collapse toward one behaviour [6, 33]. Structure announced as emergent is hard to tell from structure built into the protocol. Most simulations of agent populations are validated by believability rather than by measured outcomes against a control [26]. The order a population settles into is a property of the coupled learning of its members, and it is read from no single agent's instructions. The science of what emerges, which cannot be steered by command but only through its conditions, is therefore also the way to control it.

This paper is about the principles of social emergence that apply to communicating agents with autonomous perception: agents that each interpret signs through their own learning and never copy another's interpretation [7, 30]. This holds whether or not they arrive with a shared vocabulary and prior knowledge of roles, as a language-model agent does. Its claim is that something forms

socially, on top of whatever the agents bring: a system of roles that none were assigned, held in mutual expectations, which it measures as the coevolution of institutions. Like any emergent order, it is an invisible hand in Smith's sense [38]. Each agent reads and shows signs in its own interest alone, and a division of labour that none of them intended forms among them. From it the consumers obtained more. It does not always form, so the conditions are the finding. Earlier work in this line used other learners [12, 13]; one learner kind, an evolution strategy, is reported here, and whether the same structure forms with other learners is not tested (Section 7). The agents here are simple and know nothing of roles beforehand, so what forms can only have arisen from the interaction: the paper studies the social formation of meaning by separating it out from what a pretrained language model brings. The simplification is the method, as it is when a neural network simplifies neurons to show how cognition forms: the lower level is modelled, and still an upper level emerges that was not programmed in. The test of emergence used here, following Duong and Grefenstette [13], is that a population that starts homogeneous differentiates through interaction into a system of distinct, interlocking roles that none were assigned. It operationalizes Coleman's point [10] that the transition from individual action to social structure is not an aggregation of behaviour but an organization of interplaying actions. Three commitments shape the design. The first is *autonomous perception*: no agent is handed another agent's interpretation. Each learns for itself what a sign is worth, through its own trial and error on its own reward, so a shared meaning is an achievement of the population and not a gift of the designer. The second is *emergence*: a sign acquires a meaning because the agent that displays it and the agent that seeks it are rewarded together, without either knowing the other's reading. A structure is called emergent here only when it is measured against a control that removes the signs and against a shuffled null. The third is *social science*: the division of labour, the convention and the order held by mutual expectation are taken in the sense the social sciences gave them [16, 28, 42], and they are grown where each can be counted, so that a claim of social theory becomes a measurement. The learner is a parameter of the design; the evolution strategy was chosen so that each condition can be isolated and the arithmetic behind it checked.

The setting. The economy is a market of programs. Two kinds of agent take part. The *consumers* are eleven buyers with fixed wants. Each runs whatever program it buys on its own data and pays by how well it performs. The *learners* are sixteen agents that change what they do by trial and error on what they earn. Each learner carries a short *plan*, a list of actions of three kinds: buy a part from another agent, make a part, or sell a product. The number of lines in the plan is fixed and is the learner's *capacity*. A *product* is the program a sell line offers, built from the parts made or bought in the lines that follow it. The complete program a consumer wants is the *whole*, and the pieces it is built from are *parts*. A whole can be built by one agent that makes every part itself, or *assembled* by an agent that buys some of them. Time passes in *rounds*. In each round every agent posts its plan and its sign, every buying line draws one seller, the deliveries are run and paid, and every learner adjusts its plan and signs toward what paid (Figure 1).

The restriction of perception to signs has a basis. In the symbolic-interactionist account of how roles and institutions form, people act toward one another on the basis of the meanings they have learned to attach to one another's signs: a name, a title, a uniform, a storefront, a price, a reputation. Those meanings arise from interaction and are revised by it [7, 30]. The sign here is the abstraction of everything by which one agent recognizes what another is for. Restricting perception to it isolates the one mechanism by which meaning can arise between agents that are given none. The design generalizes signalling games, in which a signal acquires meaning because sender and receiver are rewarded together [28, 37], to the signalling of a role through a market, and tag models [36], in which tags are matched or copied, to meanings that each side learns separately. Here an agent sees nothing of another except its *displayed sign*, a vector of sixteen numbers the other has learned to show, and each buying line carries a *sought sign* of the same form. The seller is drawn at random with a weight that rises steeply with the similarity between the two (Section 3). Nothing else of another agent is observable, not its plan, its capacity, its kind or its history. Nothing passes between agents except displayed signs, delivered programs and payments, and each agent's reading of a sign is its own, shaped by its own reward alone.

The consequence is a *double induction*, Parsons's double contingency [34] made computational. A seller learns what sign to display from what sells. A buyer learns what sign to seek from what it receives. Neither induction can finish alone, since a displayed sign means nothing until seekers heed it, and a sought sign finds nothing until displayers converge on it. Each agent holds only its own half of every meaning, and the meaning exists only where the halves meet. This is what makes a structure that forms here a property of the population rather than of trained individuals, and what lets it outlive them: a newcomer does not inherit a meaning, it induces its half from the halves already in use around it. The arrangement also puts together, in the order in which they occur in human societies, learning from reward and the learning of meaning. Every agent learns only from the consequences of its own actions, and what it learns includes the meanings of signs that exist only in use among others. This is a step towards treating autonomous perception as a condition of emergence, and every measurement below depends on it: what emerged socially can be told from what the agents were given only when each agent's interpretation is its own. The control for every run is its *sign-blind twin*, the same market on the same random seed in which every seller is drawn with equal weight and nothing else is changed. Whatever differs between a run and its twin is due to the signs. That signs improve routing once they mean something is not in question, and it is not the finding. At the start they meant nothing. In every world started blank, each agent began with random signs. No vocabulary was shared, and no agent could copy another's reading. The finding is that under the stated conditions the agents developed a sign system: a seller's sign came to predict what it sells. With it they developed an institution, a system of roles held in mutual expectations that none of them was given. Where a condition was removed, as in the worlds without the tree of wants, neither formed (supplement).

A social structure, if it emerges, should be visible in three ways, each measured against the twin. Roles should appear. Wholes should be *assembled*, some parts bought from other learners rather

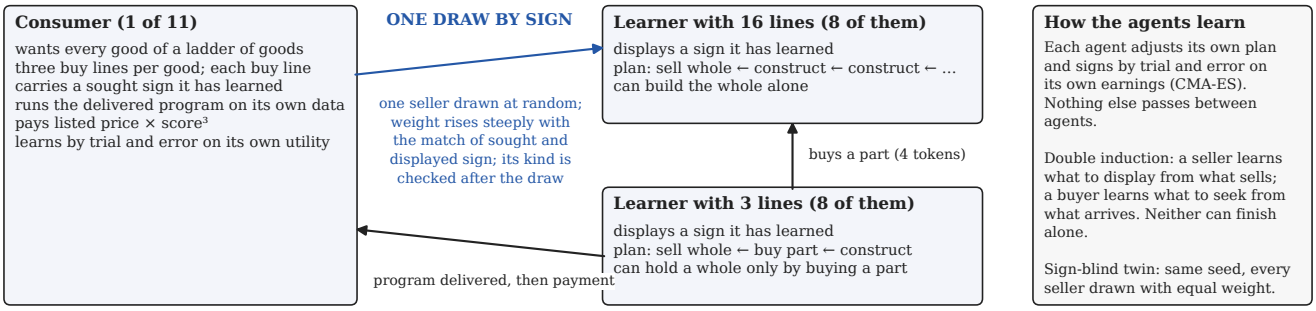


Figure 1: One round of the market. A consumer's buy line draws one seller at random, weighted steeply by the similarity of its sought sign to the seller's displayed sign, and the seller's kind is checked only after the draw. The delivered program is run on the consumer's data and paid by measured quality. A 16-line learner can build the whole alone; a 3-line learner can sell it only by buying a part. Every agent learns by trial and error on its own earnings alone. The sign-blind twin draws every seller with equal weight.

than all made by the seller. And the consumers should obtain more of what they want. A *role*, in the measurable sense used throughout, is a group of sellers selling the same kind of thing that an outsider who sees only the signs can tell apart from the other groups better than chance; it is the recognizable core of the sociologist's role, not all of it. No agent is assigned a role, and each learns what to display and what to seek only through its dealings with the others, so no role here was put in by design. What is to be shown is that roles form, that the signs and not the rules produce them, under which conditions (Section 4), and that they are held by the population rather than by its individuals.

Our contributions are:

- A market model in which buyers choose among all offers by sign alone and pay by measured quality (Section 3), with derived and checked conditions for when buying a part pays more than making it (Section 4).
- Two worlds built to those conditions, with a sign-blind twin on every seed and the measurements declared before the runs (Section 5). In them, agents that began with meaningless signs developed a sign system. The sellers sorted into roles legible from their signs in 10 of 12 machine-learning and 12 of 14 cipher worlds, against 1 and 2 of the twins. The draw by those signs routed consumers to the rare sellers and raised their utility (Section 6).
- The condition under which the division of labour becomes the assembly of wholes from bought parts. When no learner can build the whole within its capacity, every whole the consumers paid for was assembled through the sign draw; when half can, which kind of learner takes the market first decides the seed (Section 6.4).
- The test that what formed is a social structure, an order over the individuals, rather than a habit of particular agents. Under births and deaths that replaced nearly every founder, the roles and the assembling teams were held by learners born into the market (Section 6.5). Coevolutionary reconstruction reads what of that society one agent carries and what only the society holds (Section 6.6).

2 Related work

Emergent conventions and signals. Lewis's conventions [28] and their evolutionary treatment [37] explain how a signal acquires meaning when sender and receiver are rewarded together. Language games [39] and lexicon formation in neural agents [24] showed shared names arising without a designer. Deep multi-agent work learns protocols by a shared gradient [20, 27, 31]. Roles emerge in cooperative deep reinforcement learning as latent variables shared across agents [41], and bartering has been grown among reinforcement learners in a gridworld market [25]; in both the agents share an observation space the designer fixed. Tags sustain cooperation without meaning [36]. This paper continues a line in which institutions and roles emerge from agents' own interpretations of signs, each learner with its own update, and are read off a market of co-adapting agents [13, 14].

Division of labour and markets. The division of labour as the source of a society's wealth is Smith's subject [38], and as the source of its cohesion Durkheim's [16]. An order held by the expectations of those within it is Weber's legitimate order [42]. Generative social science grows a regularity from local rules before calling it explained [17, 18]. Axtell grew firms this way [3], and agent-based computational economics grows markets [40]. That an institution can arise among independently learning artificial agents was shown for money by Marimon, McGrattan and Sargent [29]. Cooperative coevolution evolves the parts of a solution in separate populations, and the decomposition can emerge as species are added and removed, but the collaboration that assembles the parts is fixed by the framework [35]. Here the decomposition, the roles and the routing emerge from one market.

Language-model agents, and their validation. Such agents simulate social behaviour [9, 32, 33], scale to populations [8] and form conventions [2], though whether those conventions are formed in interaction or carried in from training is disputed [4]. They are organized into teams by designed roles and protocols [11, 23, 43]. Their promise as a research method is argued in [1]. A review

finds most generative agent-based models validated by believability rather than by measured outcomes [26].

3 The model

Agents. A world has 16 learners and 11 consumers. Every agent is a single vector of numbers between 0 and 1, its *genome*. The genome is read off into the sign the agent displays, 16 numbers, and into its plan. A learner's plan is a list of lines whose length, its *capacity*, is fixed by design at 2, 3 or 16 and is invisible to the other agents. Each line has a kind, a sought sign and an item. The kinds are *stop*, which ends the plan, drawn with probability 0.1, and *buy*, *construct* (make a part) and *sell*, drawn with probability 0.3 each. A buy line also carries the test and data by which the bought part is judged. A consumer's plan has buy lines only, three for each good it wants (Section 5).

Items are names. The things an agent can make, buy or sell form a tree of names with four levels, family, kind, class and leaf, a leaf being one concrete part. An item is one number per level, the lowest tenth of each range meaning *stop here* and the rest shared among that level's children, so names close in the tree are close as numbers and a small change of the genome moves an agent to a neighbouring item. A buy line that stops early asks for anything in the category it names; a sell line that stops early sells that category's designated default, the choice the world's goods most often need, common but wrong for the rest.

Products. A product is what one sell line offers. It consists of the sell line and the construct lines that follow it up to the next sell or stop, nested in order with the sell line outermost. A buy line inside that stretch splices in the product of the supplier the line drew. So [sell classifier, construct labeler, construct vectoriser] is the program classifier(labeler(vectoriser(text))), run on the buyer's data.

One draw by sign. Every sell line in the world is an offer. For each buy line one offer is drawn at random with weight $(c - c_{\min})^4$. Here c is the cosine similarity between the line's sought sign and the seller's displayed sign, and $c_{\min}$ is the lowest such similarity among all offers. Only after the draw is the offer's item compared with the category the line names. If it does not lie in that category, the line trades nothing this round.

Payment. Each good has a listed price. A part bought by a learner from a learner costs 4 tokens. A consumer runs the program it receives on its own data with its own test. It pays the listed price times the cube of the score. An incomplete whole scores 0 and earns nothing. A consumer's fitness is a Cobb–Douglas utility over the qualities of the goods it obtains. This product rewards obtaining some of every good, so a want that brings nothing loses ground to the others. A learner's fitness is the tokens it earns minus the tokens it pays its suppliers. A supplier is paid on every sale of a product its part is in.

Learning. Each agent adjusts its own genome by trial and error on its earnings. It uses its own copy of the evolution strategy CMA-ES [21, 22] with the full covariance update and an initial step size of 0.1. One candidate is tried per round. This is learning from reward in the sense of reinforcement learning, each agent learning only from what its own actions earn, carried out by an evolution strategy rather than by a gradient. Because every agent's earnings depend on what the others currently do, the agents' searches coevolve: the market is a coevolutionary system. Plans are read from genomes in the manner of gene-expression programs [19].

4 Conditions

Whether buying a part pays more than making it is a matter of two rates. Consider a learner that tries one candidate per round around the centre of its search with step size σ. It stays on a part of width w with probability $s = \operatorname{erf}(w/2\sqrt{2}\,\sigma)$. A good of k parts is right only when every part is. A lone builder is therefore right on a share $r_{\text{alone}} = \prod_i s_i$ of rounds. An assembler is right on $r_{\text{asm}} = \delta \prod_i s_i$, where δ is the chance that its draw finds the right supplier. Let P_w be the price of the whole and P_p the price paid for parts. Assembling is selected over building alone only if

$$r_{\text{asm}}\,(P_w - P_p) > r_{\text{alone}}\,P_w. \tag{1}$$

So where every part is easy to stay on and a learner has capacity enough for the whole, building alone is selected whatever the signs do. Buying is selected only where the whole needs more lines than the learner has, or where a lone builder rarely gets the parts right together. The worlds below therefore make the whole hard to find. Each class has several keys, and the whole needs the right key in every class: this is the *hard key*. The whole is *all-or-nothing*, paid only when complete. Replacing a two-class product by the assembled whole then pays only when the chance of drawing the right supplier times the chance that the supplier's half is right exceeds the price of the two-class good divided by the price of the whole: 0.125 in the cipher world and 0.25 in the machine-learning world at the list prices below (supplement). Consumers want every rung of a ladder of goods, because a learner's search carries information about a good only if some candidates are paid for it. A lone builder can climb one rung at a time, and the question is whether, under hard keys, it reaches the whole before an assembler does (supplement).

5 Worlds and measures

The cipher world. The goods are decoders for enciphered text. Each letter belongs to one of four position classes, its position in the word modulo four. Each class is enciphered with one of four substitution keys. A part is a decoder that undoes one key on one class. The whole decodes all four classes. Consumers want seven goods: the four texts with one class decoded, the two with two classes decoded, and the whole. Their listed prices are 10, 30 and 240 (a second version prices the whole at 120). A class counts as decoded only when at least 99% of its letters come out right, so a wrong key earns nothing by chance. A one- or two-class good scores the share of its classes decoded. The whole scores 1 when complete and 0 otherwise. Of the 16 learners, 8 have plans of 3 lines and 8 have plans of 16. A 3-line learner can deliver the whole only by buying a part. A 16-line learner can build it alone. Each of the 11 consumers wants all seven goods, with utility $U = \prod_g (0.3 + q_g)^{1/7}$ over the qualities q_g it obtains, and holds three buy lines per good. A line's good is read as a tree, first the rung and then the good within it, so a small step in the genome moves a line to

a neighbouring good. The line seeks the sign of the consumer's standing want for that good. The control for this reading is the same world with the good read from one number cut into seven slices (*without the tree*). In configuration A a learner's test names a category of data, and a bought part is scored as the mean over the category's corpora. A right part is then paid a few hundredths of its price: the part trade routes and assembles but carries almost no tokens. In configuration B the test names the buyer's own corpus, so a right part scores 1 and is paid 4. Every world has a sign-blind twin on the same seed, in which every offer is drawn with equal weight and nothing else changes. Worlds run 40,000 rounds, and every tenth round is logged. The measurements are taken from the record cut at rounds 10,000, 20,000 and 30,000, and from the whole record at the end.

The machine-learning world. There are two corpora, movie-review sentiment and fake-news detection, with 1,000 documents each. Each corpus has three rungs: an embedding, a labelled embedding (an embedding with the corpus's labels attached) and a whole classifier. Each of the 11 consumers wants every rung of both corpora, six goods, at list prices 30, 60 and 240 times the delivered score cubed. Its lines are read as a tree over the rungs, as in the cipher world. A part bought from a learner is judged by the test and data the buyer's line names and is paid 4 times the score cubed. In the main set 8 learners have plans of 2 lines and 8 have plans of 16. A whole built alone takes three lines. A 2-line learner can therefore sell a whole or a labelled embedding only by buying the rung below it, while a 16-line learner can build everything alone. The *capacity arm* changes only the plan lengths, to 8 learners with 2 lines and 8 with 3. Every world has a sign-blind twin. The sets are given with the results.

Birth and death. In the birth-death versions learners die and newborns arrive with fresh genomes. No sign, want or practice is inherited (rates in Section 6.5).

Reconstruction. In coevolutionary reconstruction a fresh world is seeded with statistics measured on a mature one. From a source run's last 2,000 rounds, two things are measured for each learner: its *practice*, the lines of its first product as (kind, item) pairs, and the share of learner-rounds that practice holds among the learners of the same capacity. The practices are dealt to the new world's learners in those proportions, with no price, test or sought sign. There are four levels. *Presence* deals the practice and its display-sign centroid. *Outward* deals the practice alone. *Half* deals the practice to every other learner. *Keyless* deals the practice with every decoder key redrawn at random, which carries the division of labour and nothing of the solution to the keys; it is run at the blank learners' step size.

Measures. Four things are measured, each from the record alone. The first is *what was delivered, and who made it.* From the offer the buyer chose, the delivered program is reconstructed by following every bought part to the supplier's line that made it, and every part that supplier bought in turn. A delivery is *assembled* when at least two learners each supplied a part the good requires (a decoder of the right class and key; a rung). It is *fully right* in the cipher world when it scores at least 0.99. It is *scoring* in the machine-learning world when its F1 score (a classifier's accuracy measure) is above

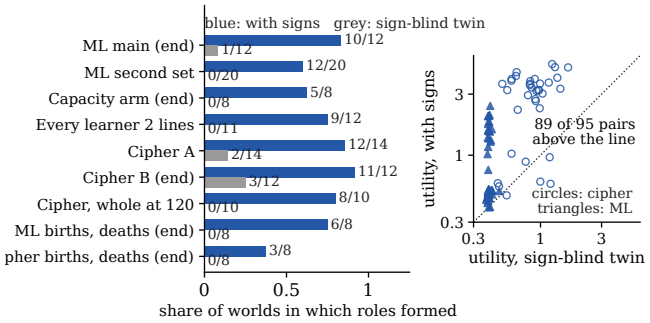


Figure 2: Every set, the world with signs against its sign-blind twin. Left: the share of worlds in which roles formed, that is, an outsider could tell a seller's product from the other sellers' signs on both accuracy and mutual information, with the counts; rows marked (end) are read at the end of the runs, the others at round 20,000. Right: consumers' mean utility with signs against the same seed's twin, one point per pair; above the dotted line, utility was higher with signs.

0. The second is *roles legible from the signs.* Within a window of 100 rounds each (seller, product) pair is a unit. A unit's product is guessed as the product whose average sign over the *other* sellers' units is nearest to this seller's sign by cosine. A product sold by one seller alone counts as a miss, so a sign that only marks one individual earns nothing. The accuracy of the guesses and the mutual information between true and guessed products are compared with the same quantities after the products are shuffled among the window's units (200 shuffles, smallest attainable p 0.005). Roles have *formed* in a world when both quantities exceed their shuffled values at $p < 0.05$. The guess is made by the product and separately by the head item alone. The third is *trade between learners.* It is counted only in tokens that moved: the share of consumer-paid deliveries whose seller's product contains a part bought from another learner and paid for in that delivery. The fourth is *what the consumers got*, their mean Cobb–Douglas utility per window of 1,000 rounds, with 0.3 when a consumer obtains nothing. *Concentration* is the Gini coefficient of the learners' income from consumers over the last 200 logged rounds, and the top earner's share of it. Every read file behind a count in this paper, the engine and the world generators are in the supplementary material.

6 Results

Telling emergence from design. Four instruments do this (Figure 2 and Table 1 give every set). The *sign-blind twin* is the same market on the same seed with sellers drawn at random, so whatever appears with signs and not in the twin was not put there by the rules. The *shuffled null* tests the roles read against the same statistic with products shuffled among the sellers, so a role that only reflects common products scores nothing. *Births and deaths* replace nearly every founder, so a structure that persists is held

Table 1: Every set, with signs against the sign-blind twin. Roles: worlds whose sellers formed roles by product (Section 5), signs vs twins, Fisher's one-sided p. Utility: pairs with higher consumer utility with signs, one-sided sign-test p. Wholes: machine-learning sets, worlds with any scoring whole, signs vs twins (Fisher's p); cipher sets, pairs with more right wholes with signs (sign-test p). End: the whole record at the run's end; otherwise the record cut at the round named.

Set, pairs, read	Roles (p)	Utility (p)	Wholes (p)
ML main, 12, end	10/12 vs 1/12 (3e-4)	12 of 12 (2e-4)	sold 5/12 vs 1/12 (0.077)
ML second set, 20, 20,000	12/20 vs 0/20 (2e-5)	20 of 20 (1e-6)	sold 9/20 vs 3/20 (0.041)
Capacity arm, 8, end	5/8 vs 0/8 (0.013)	8 of 8 (0.004)	sold 4/8 vs 0/8 (0.038)
Every learner 2 lines, 12, end	9/12 vs 2/11 (0.009)	11 of 12 (0.003)	sold 4/12 vs 1/12 (0.158)
Cipher A, 14, 20,000	12/14 vs 2/14 (2e-4)	11 of 13 (0.011)	up 12/14 (0.006)
Cipher B, 12, end	11/12 vs 3/12 (0.001)	8 of 8 (0.004)	up 12/12 (2e-4)
Cipher, whole at 120, 10, 20,000	8/10 vs 0/10 (4e-4)	9 of 10 (0.011)	up 8/10 (0.055)
ML births and deaths, 8, end	6/8 vs 0/8 (0.003)	8 of 8 (0.004)	sold 3/8 vs 1/8 (0.285)
Cipher births and deaths, 8, end	3/8 vs 0/8 (0.100)	3 of 4 (0.312)	up 4/8 (0.637)

by the population. *Reconstruction* deals a mature population's practices to a fresh one: what comes back by itself is what one agent carries, and what has to be coevolved again is what the society holds. Paired counts carry a one-sided sign test and counts of worlds a one-sided Fisher's exact test against the twins, in the direction declared before the runs (signs above twin); the roles read carries its shuffled null. A milestone read covers the 2,000 rounds before it, from the record cut there; the end read covers rounds 38,000 to 39,999.

6.1 What the consumers obtained, and where the buyers were routed

The machine-learning world's main set is 12 seed pairs run to 40,000 rounds (per-seed table in the supplement). The consumers' utility in the last window was higher with signs than blind on 12 of 12 pairs (sign test $p = 2.4e-04$), 0.52 to 2.06 with signs and 0.39 to 0.42 blind, where 0.30 is a consumer that obtains nothing. The gain came through routing. Labelled embeddings, which few learners sell, were bought and scored by consumers more often with signs on 12 of 12 pairs ($p = 2.4e-04$). Whole classifiers were sold and scored in 5 of 12 worlds with signs and in 1 of 12 twins ($p = 0.077$). A second set of 20 pairs of the same design, the blank starts of Section 6.6, gave the same picture at 10,000 and 20,000 (Table 1), with labelled embeddings higher on 18 of 20 ($p = 0.0002$). Over the 32 pairs of the design, wholes were sold in 14 worlds with signs and 5 twins at 20,000 ($p = 0.014$). The signs do not create the whole-seller on most seeds; they raise the chance of one and, as Section 6.3 shows, decide what kind it is.

In the cipher world's configuration A, 14 seed pairs ran at the list price of 240. Fully right wholes delivered to consumers in the window were more numerous with signs on 12 of 14 pairs at 10,000 and at 20,000 rounds ($p = 0.0065$), and on 14 of 14 at the stop ($p = 0.0001$). (The worlds were stopped between rounds 24,000 and 31,000 to free the machines.) The factors ranged from about 1.1 to about 1,400 at 10,000 (per-seed counts in the supplement). Consumers' utility was higher with signs on 12 of 13 pairs read at 10,000 ($p = 0.0017$) and on 11 of 13 at 20,000 ($p = 0.011$), 0.60 to 5.16 against 0.55 to 1.64.[1] Under configuration B a bought part is paid its listed price. There right wholes were higher with signs on 7 of 12 pairs at 10,000 and utility on 8 of 12, neither reaching $p < 0.05$. At 20,000 both were higher on 11 of 12 ($p = 0.0032$). At the end both were higher on 12 of 12, and on the 8 pairs read (Table 1) utility was 3.03 to 4.03 against 0.50 to 1.35.

6.2 The emergent institution: roles legible from the signs

Roles have formed in a world when an outsider who sees only the other sellers' signs can tell what a seller sells, with both the accuracy and the mutual information of the guess above their shuffled values at $p < 0.05$ (Section 5). In the machine-learning main set roles formed by product in 10 of 12 worlds with signs against 1 of 12 twins at the end (Fisher's exact test $p = 3.2e-04$). Accuracy alone exceeded its shuffled value in 12 of 12 worlds with signs and 3 of 12 twins, since a twin can guess above chance from the frequencies of products alone; that is why mutual information is required too. Table 1 gives every set: roles formed in most worlds with signs and in almost no twin.

In configuration A roles formed by product in 7 of 14 worlds with signs against 1 of 14 twins at 10,000 ($p = 0.016$), and in 12 of 14 against 2 of 14 at 20,000 ($p = 0.0002$). At the stop they formed in 11 of 14 against 0 of 14 ($p = 1.7e-05$). Accuracy alone was above its null in 14 of 14 worlds with signs and in up to 6 of 14 twins, so again both tests are needed. Under configuration B roles formed by product in 10 of 12 worlds with signs against 0 of 12 twins at 10,000 ($p < 10^{-4}$), and in 11 of 12 against 3 of 12 at the end. With the whole priced at 120 they formed in 8 of 10 against 0 of 10 at 20,000 (Table 1).

6.3 Market structure: two regimes from the same rules

Where a whole-seller emerged, the learners' income from consumers concentrated on it. In the five whole-selling worlds of the machine-learning main set the income Gini at the end was 0.76 to 0.90, and the top earner held 0.37 to 0.93 of all learners' income. The twins had a Gini of 0.04 to 0.17, and the worlds with signs that sold no whole had 0.36 to 0.61. The five are of two kinds. In 516, 518 and 714 one 16-line learner built the wholes with its own construct lines and held 0.92 to 0.93 of the income: a monopoly of a lone builder. In 517 every whole was assembled from a labelled embedding bought from another learner, and in 515 about a third were, with the top earner at 0.37: a team. The cipher world shows the same two regimes sorted by capacity. At 20,000 in configuration A, on seeds 316 and 324 16-line learners built nearly every

[1] Uneven denominators: the world with signs on seed 312 has no utility line in its read at 10,000; the roles read of the twin on seed 309 in the 2-line arm returned no result at 20,000, nor the twin on seed 308 at the end; two twin runs on seeds 310 and 311 in configuration A, which have no world with signs, are excluded.

whole alone. On seeds 315, 321 and 322 the wholes were sold by 3-line learners, and every one was assembled, since a 3-line learner cannot hold a whole without buying a part (per-seed counts in the supplement). The same rules produce a monopoly on one seed and a distributed market with supply chains on another. Which it is was decided early, by which kind of learner took the whole's market first. Assembly exists in the twins too: (5,723/18,799) of their right wholes in configuration A at 20,000, against (97,795/184,246) with signs. So assembling as such comes from the hard keys and the ladder of goods. The signs set its scale, about twenty times the wholes, and make a supply chain that holds.

6.4 The capacity arm: when no learner can build the whole alone

The capacity arm (Section 5) leaves no learner able to build a whole without spending its entire plan on it or buying the rung below it; eight seed pairs were run.

At 10,000 and 20,000. Consumers scored wholes in 4 of the 8 worlds with signs (716, 321, 322 and 323) at both milestones, and in none of the twins beyond a single whole draw at 10,000, (1/3,637) on seed 320 ($p = 0.14$ counting that draw, 0.038 at 20,000). In 716 and 321 every whole was assembled from a bought labelled embedding; in 322 and 323 one 3-line learner spent its entire plan on the whole and built it alone (supplement). Utility was higher with signs on 8 of 8 pairs at both milestones ($p = 0.0039$, the smallest attainable value at this n), and roles formed by product in 7 of 8 and then 6 of 8 worlds with signs against 0 of 8 twins.

At the end. All eight pairs ran to 40,000 rounds. In the last 2,000 rounds consumers scored wholes in the same 4 of 8 worlds with signs and in 0 of 8 twins ($p = 0.038$): on (20,767/29,961), (24,791/29,635), (18,672/21,726) and (15,304/29,988) of their whole draws in 321, 322, 323 and 716. In 321 and 716 every whole was still assembled, and (20,793/28,296) and (15,304/22,605) of all consumer-paid deliveries in those two worlds paid a supplier, against (7/22,805) and (0/37,249) in their twins. Utility and roles at the end are in Table 1.

Every learner with 2 lines. A further arm gives every learner 2 lines, so that no learner can build a whole without buying the rung below it (12 seed pairs, 306 to 317). At 10,000 consumers scored wholes in 4 of 12 worlds with signs and in 0 of 12 twins ($p = 0.047$), every whole assembled, (19,884/19,884) pooled; with 2 lines no learner can build a whole, so every whole on either side is assembled. At 20,000 and at the end the same 4 of 12 sold wholes, against one twin ($p = 0.16$). Roles formed by product in 8 of 12 against 0 of 12 already at 10,000 ($p = 0.0007$), and utility was higher with signs on 10 of 12 at 10,000 and 20,000 ($p = 0.019$); the end is in Table 1.

6.5 An institution, not a habit: what outlives the individuals

A social structure is an order that exists over and above the individuals who hold it and outlasts them. Whether what formed here is one, or only a habit of the particular learners that formed it, is tested by replacing them. A learner dies at 0.000125 per round, a mean life of 8,000 rounds. Newborns arrive from the broad prior at 0.004 per round up to a population of 16, inheriting no sign, want or practice. About 1.3 of the 16 founders remain after 20,000 rounds and 0.1 after 40,000. In the machine-learning world (8 seed pairs, 306 to 309 and 412 to 415) roles formed by product in 6 of 8 worlds with signs against 0 of 8 twins at 20,000 and at the end, and by head in 7 of 8 against 0 of 8 at the end ($p = 0.0007$). Utility was higher with signs on 8 of 8 pairs at both ($p = 0.0039$, the smallest attainable at this n). Two of the eight worlds with signs, 306 and 414, grew a whole-assembling team among learners born into the market, with nearly every whole assembled at the end, (3,380/3,400) and (2,977/3,000) on the two corpora in 306. Neither of their twins scored a whole. In the cipher world (8 seed pairs of configuration B) roles formed by product in 6 of 8 worlds with signs against 0 of 8 twins at 10,000 and 20,000 ($p = 0.0035$). By head they formed in 7 of 8 against 0 of 8 at 10,000 and at the end ($p = 0.0007$). Utility was higher with signs on 8 of 8 pairs at 10,000 ($p = 0.0039$). It was higher on only 5 of 8 at 20,000 and on 3 of 4 read at the end, when roles by product had fallen to 3 of 8 (Table 1; the first four pairs end at their stop near round 24,000). Every right whole the worlds with signs delivered at 20,000 was assembled, (1,358/1,358), against (783/2,471) in the twins, which delivered more wholes under turnover; at the end, (1,719/2,715) against (533/2,355). The roles, the routing and the assembling teams were held by learners who were not there when they formed. This is social structure in Durkheim's sense [15], and the objectivation Berger and Luckmann describe [5]: it is external to any one agent, it outlives the agents' lifetimes, and a newcomer finds it already there and takes a place in it. The knowledge is in the signs and in what the agents expect of one another, not in any agent.

6.6 What the society carries: coevolutionary reconstruction

Reconstruction deals a mature world's practices to a fresh population (Section 5). It reads what comes back by itself and what has to be learned again together. In the machine-learning world 714 one 16-line learner built every whole alone and held 0.93 of the learners' income. Its practices were dealt outward to five fresh worlds, read beside the same five seeds started blank. At 10,000 every seeded world had rebuilt the monopoly, with the top earner at 0.89 to 0.93 of the income. Of the blank starts one, seed 307, sold wholes and four sold none: 5 of 5 against 1 of 5, Fisher's exact test $p = 6/252 = 0.024$, and the same at 20,000 and at the end. A lone builder's practice is its recipe, and it replays: (134/129,163) of the seeded worlds' wholes at the end were assembled. In the team world 517 every whole was assembled. Its practices were dealt outward to 18 fresh worlds and to 2 at the half level, beside 20 blank starts on the same seeds. At 10,000, 17 of the 18 seeded worlds sold wholes, every one assembled, (62,287/62,287) pooled, against 7 of the 20 blank starts ($p = 1.5e-04$), of whose wholes (12,191/58,783) were assembled. At the end, 15 of 18 seeded worlds sold wholes, still every one assembled, (277,995/277,995) pooled, against 6 of 18 blank starts on the same seeds ($p = 0.003$), of whose wholes (26,369/129,072) were assembled; roles had formed in 17 of 18 seeded worlds. A team's practices carry its structure, where a lone builder's carry its recipe.

In the cipher world the sources were two monopolies, 320 and 322, in which one learner assembled every whole and held 0.89 and 0.95 of the income, and two distributed worlds, 317 and 324, in which 16-line learners built wholes alone. Seeded outward, the distributed sources replayed at once, with 28,836 and 32,140 right wholes at 10,000 and nearly none assembled. The monopoly sources did not replay (0 and 0 at 10,000), while their roles came back. Seeded by halves, the other half coevolved into the suppliers, with (11,934/11,934) and (15,335/15,589) of the right wholes at 20,000 assembled. The *keyless* level carries the roles and nothing of the solution. Three worlds from the monopoly source delivered 269 to 1,456 right wholes at 10,000 and 2,951 to 7,663 at 20,000, every one assembled, where the same roles with their keys had delivered none. Three from the distributed source stayed at a blank start's scale, so the keys were most of what that source's outward seed carried. A powered set ran five outward seeds and five blanks per source. All five from the distributed source replayed, 29,841 to 31,691 right wholes against 1 to 4,752 in the blanks (sign test $p = 0.031$, the smallest attainable at this n). From the monopoly source the assembling structure came back in three of five, (20,269/20,269), (25,366/25,366) and (434/434) assembled at 10,000, and not at all in two. Following coevolutionary reconstruction as proposed by Duong et al. [12], we read the gap between the seeded shares and the shares the coevolved society sustains as the reconstruction distance. It is small for what one agent does alone and larger for what agents do together. The half-seeded and keyless worlds rebuilt the latter through learners never told what to do.

7 What this does not show

Scale and generality. Every world has 16 learners and 11 consumers, plans of 2 to 16 lines and one or two corpora of 1,000 documents. Whether the conditions hold at larger scale is not tested. There is one learner kind, CMA-ES, and two program-building worlds; other learners, language models among them, are not tested.

Where the counts are small. Configuration B's first set and the cipher births-and-deaths first set are four pairs each, on which a sign test cannot reach $p < 0.05$. Their claims rest on the twelve and eight pairs with the second sets, read at 10,000 and 20,000 and, for configuration B, at the end. Under cipher turnover the utility advantage held on 5 of 8 pairs at 20,000 and 3 of 4 at the end, and roles by product in 3 of 8 at the end. The whole-seller count in the machine-learning main design reaches $p = 0.014$ only over the 32 pairs at 20,000, and it is $p = 0.077$ on the 12 pairs at the end. No correction for multiple comparisons is applied across sets and milestones; every count is reported so that the reader can apply one.

Reconstruction. The cipher reconstructions from single sources have one world per source and level, and describe rather than test; the powered set has five seeds per source.

Mechanism. The sign-blind twin was run for every comparison reported, so the differences are attributable to the signs. The roles read leaves the seller itself out, so individual recognition is not excluded from the market. With prices learned as bands instead of listed (8 pairs, supplement) almost no wholes were assembled, while roles still formed in 4 of 8 worlds with signs against 0 of 8 twins.

The criterion, and the reads. The conjunction of accuracy and mutual information was adopted after the sign-blind twins were seen, because accuracy alone exceeded its shuffled value in up to 6 of 14 twins (each alone in the supplement). Every milestone read is computed from the record cut at the milestone (supplement).

8 Discussion

What formed in these markets is an institution in the sense the social sciences give the word. It is a system of roles held in place by what the agents came to expect of one another. The roles were learned, not assigned, and they did not form in the twins. They were still there after nearly every agent who formed them had been replaced.

The conditions separate. The hard keys and the all-or-nothing whole make the whole worth more than a lone builder can climb to by partial credit, and they leave the 3-line learners no way to it but buying. Where a 16-line learner has lines enough, Eq. 1 says lone builders hold what assemblers would buy, and on most seeds they did. The consumers' wants read as a tree keep the demand for each good steady. Without the tree, assembly appeared with and without signs, and no roles formed (supplement).

The same rules produced two market structures: a lone builder's monopoly holding 0.9 of the learners' income on some seeds, and an assembling team with spread income on others. Which one formed was decided by capacity and by which kind of learner took the whole's market first, and by nothing the agents were told.

For populations of language-model agents, whose members live as long as a context window and are started and stopped continually, the result that matters is that the structure outlived its holders. An order held in the mutual expectations of a population cannot be read from any one prompt or reset by resetting an agent. Language-model agents arrive pretrained in a human language and with their roles assigned in prompts. Conventions have been reported to form among them in interaction [2], though whether they are formed there or carried in from training is disputed [4]. The conditions and measurements here apply to them as to any communicating agents with autonomous perception: the same market, the same twin and the same reads, with the learner replaced, measure what forms socially on top of what the agents bring. The design may also offer a way to train such agents socially and in a distributed manner, with goals distributed over roles as language already is, so that an agent learning from reward has contextual goals rather than one fixed objective to maximize.

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